

ECEN 607: Advanced Analog Circuit Tech

Lab 4: Op-Amp Design 1

Characterization Lab Report

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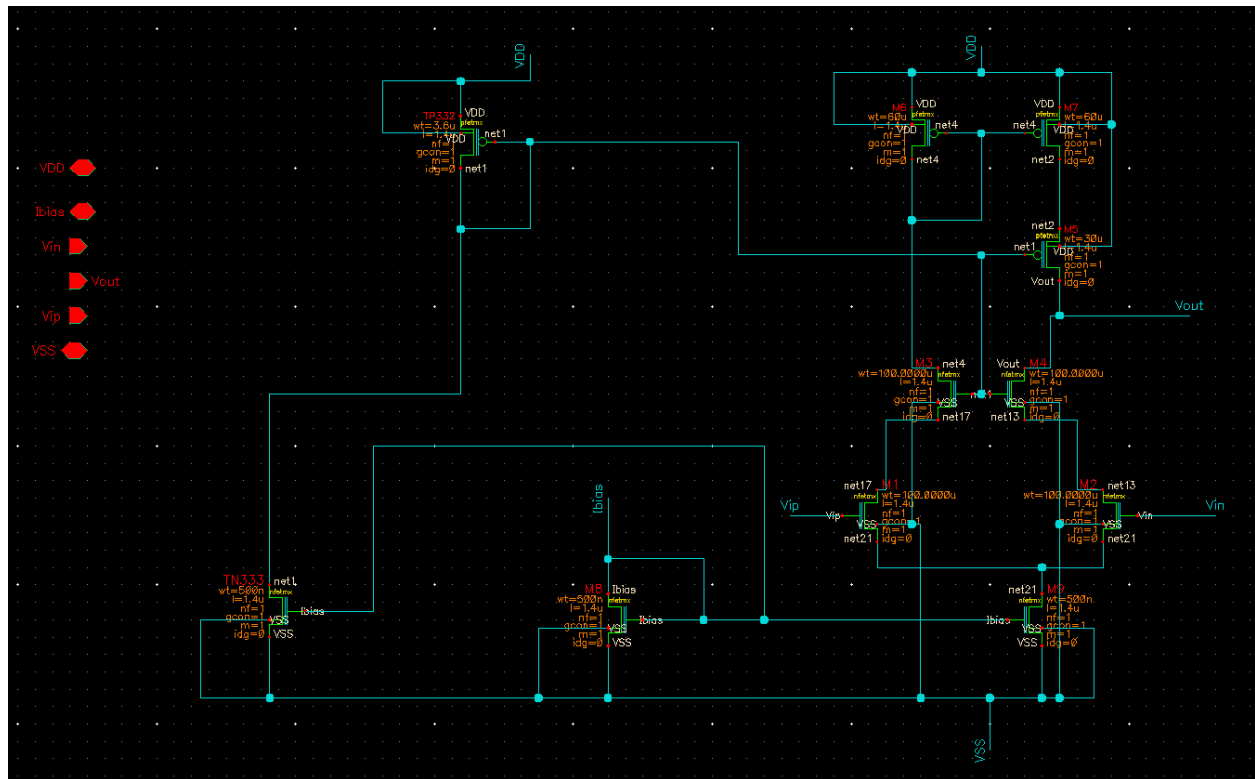
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Design:

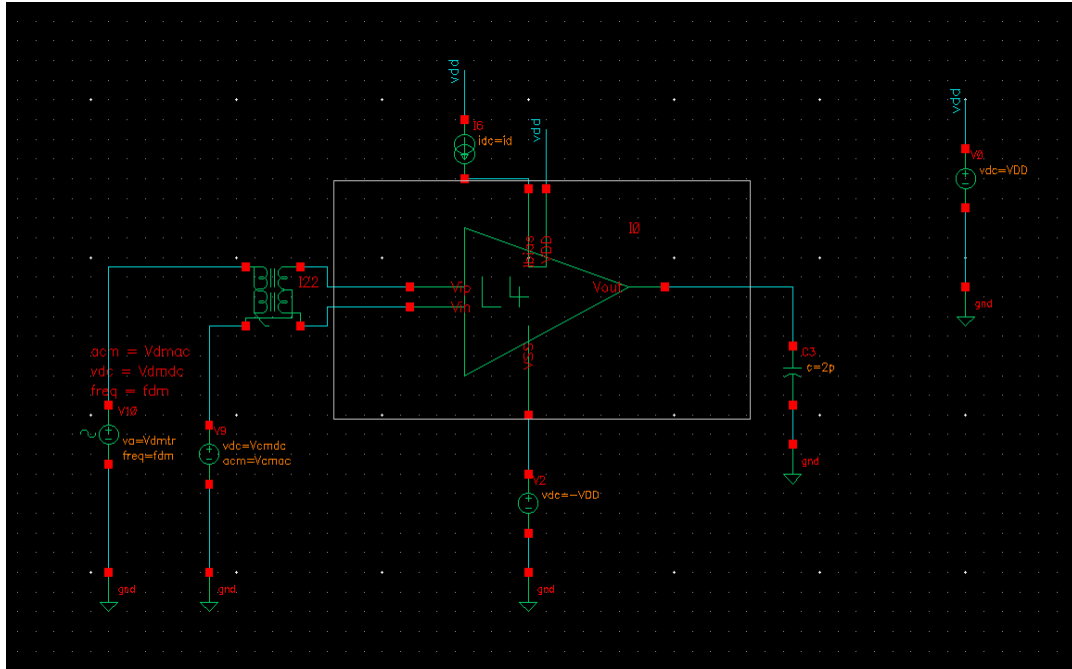
1. Simulate the circuit from the prelab and adjust the transistor sizes accordingly until all specifications are met. Provide necessary plots such that you clearly show the specs are met. Specs met and the design Parameters:

Design Parameters	Values
VB1	800mV
VB2	800mV
IB	50u
COMMON MODE VOLTAGE	0
OUTPUT SWING	1.46999

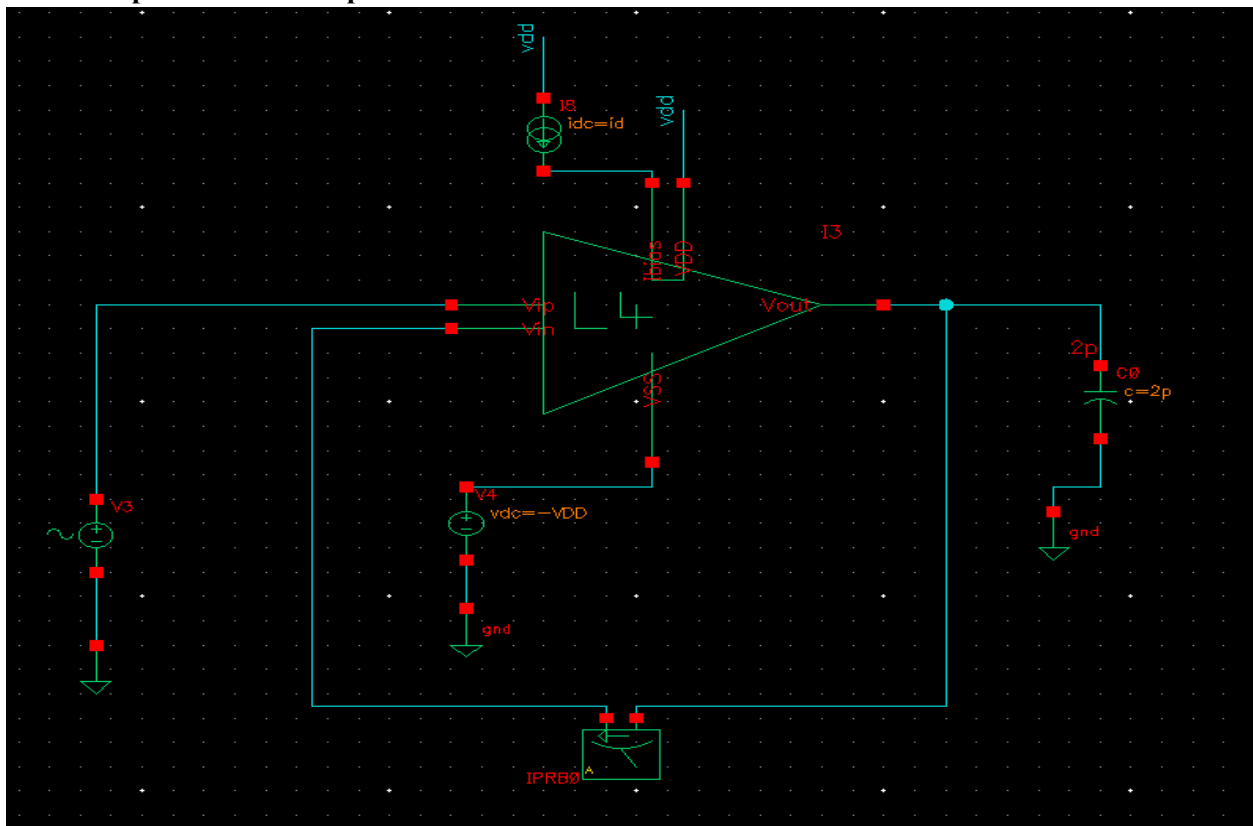
Transistor-level Schematic Screenshot:



Open-Loop Test bench set-up:

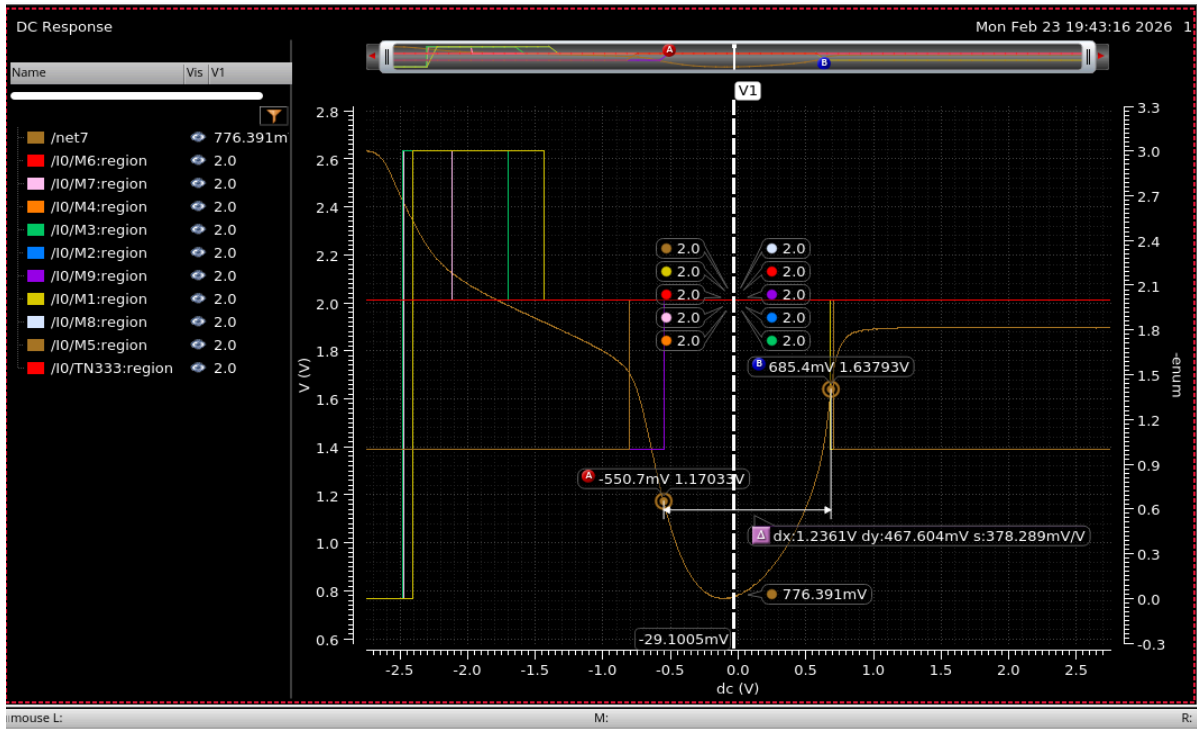


Closed loop Test bench setup:

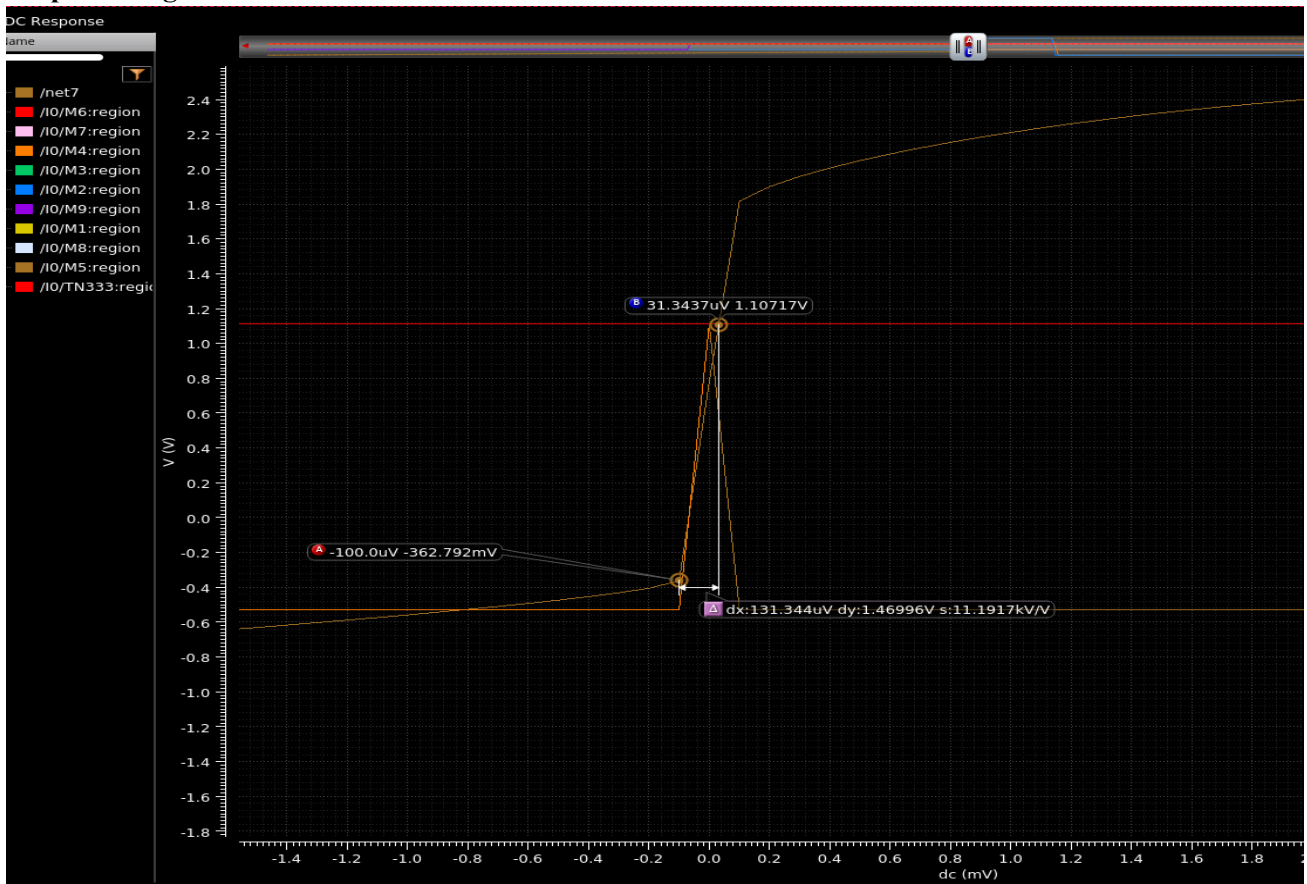


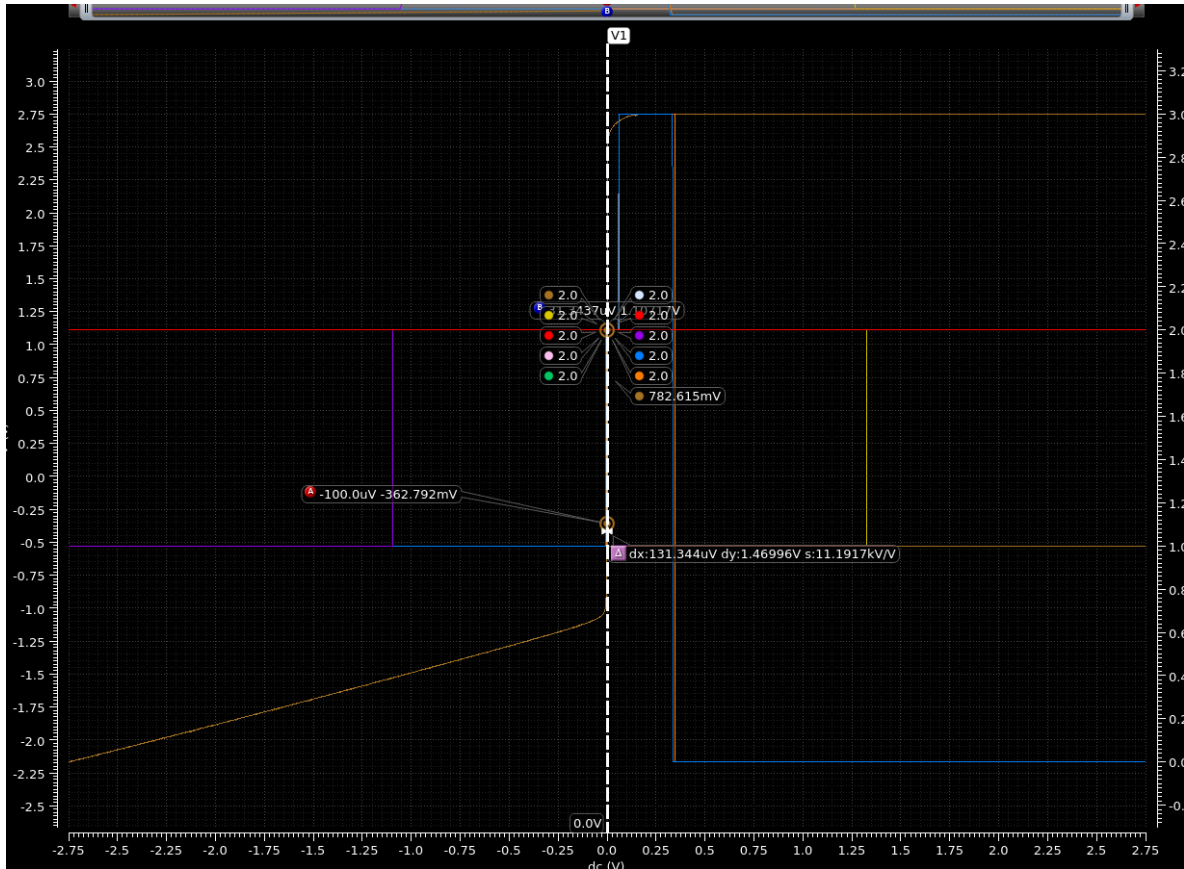
Simulation on getting proper Specifications being met:

Dc simulation and the operating point:

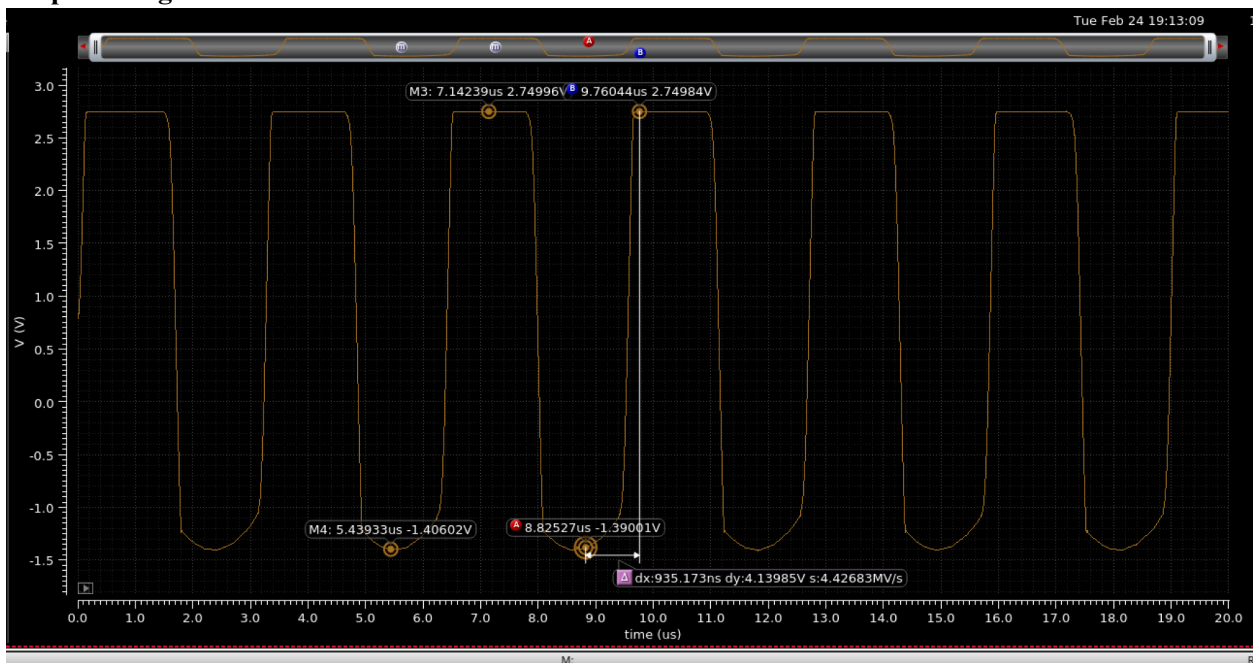


Output Swing:

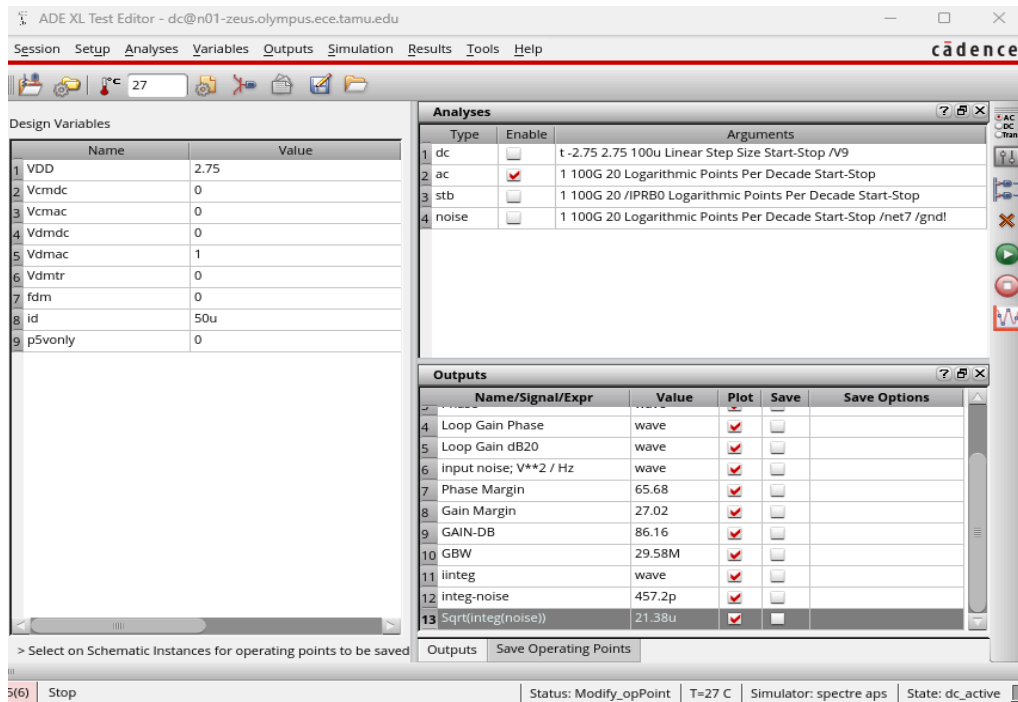




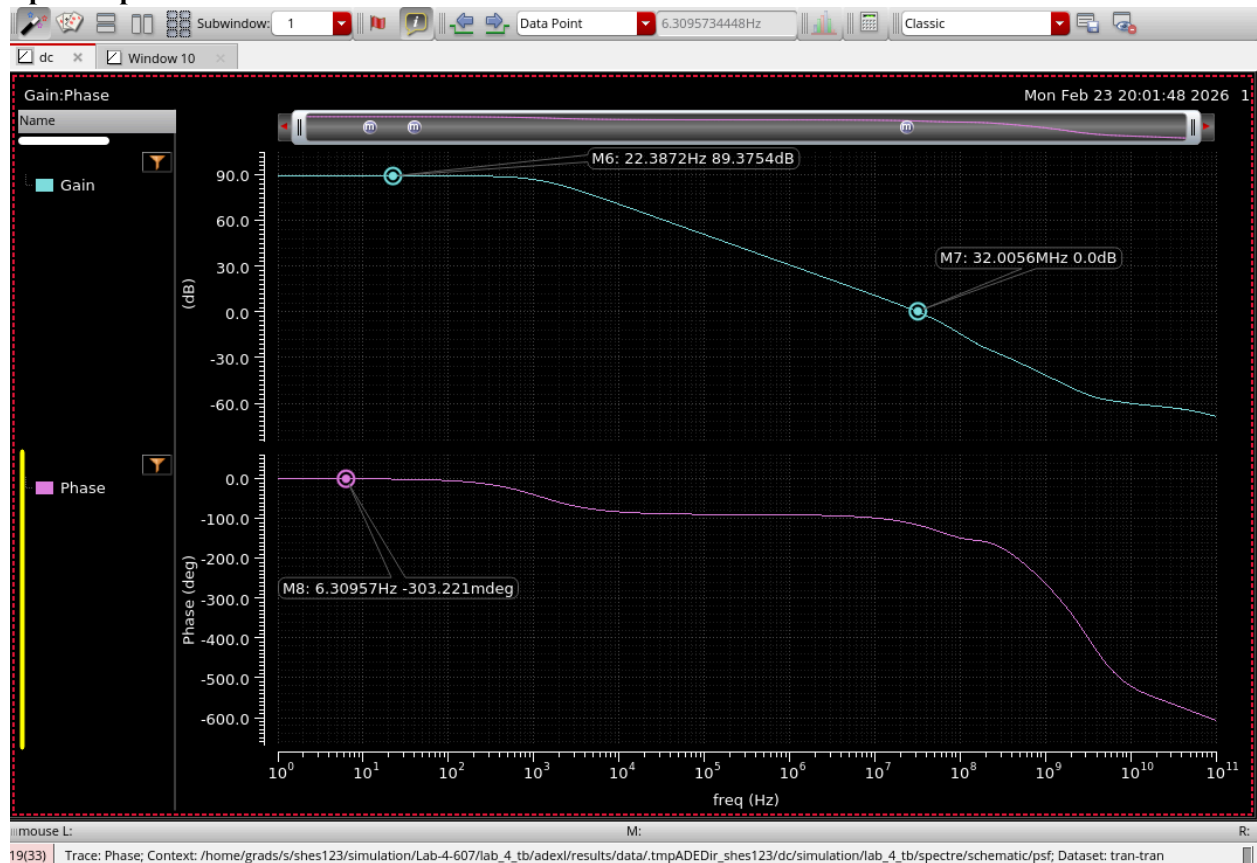
Output swing Transient:



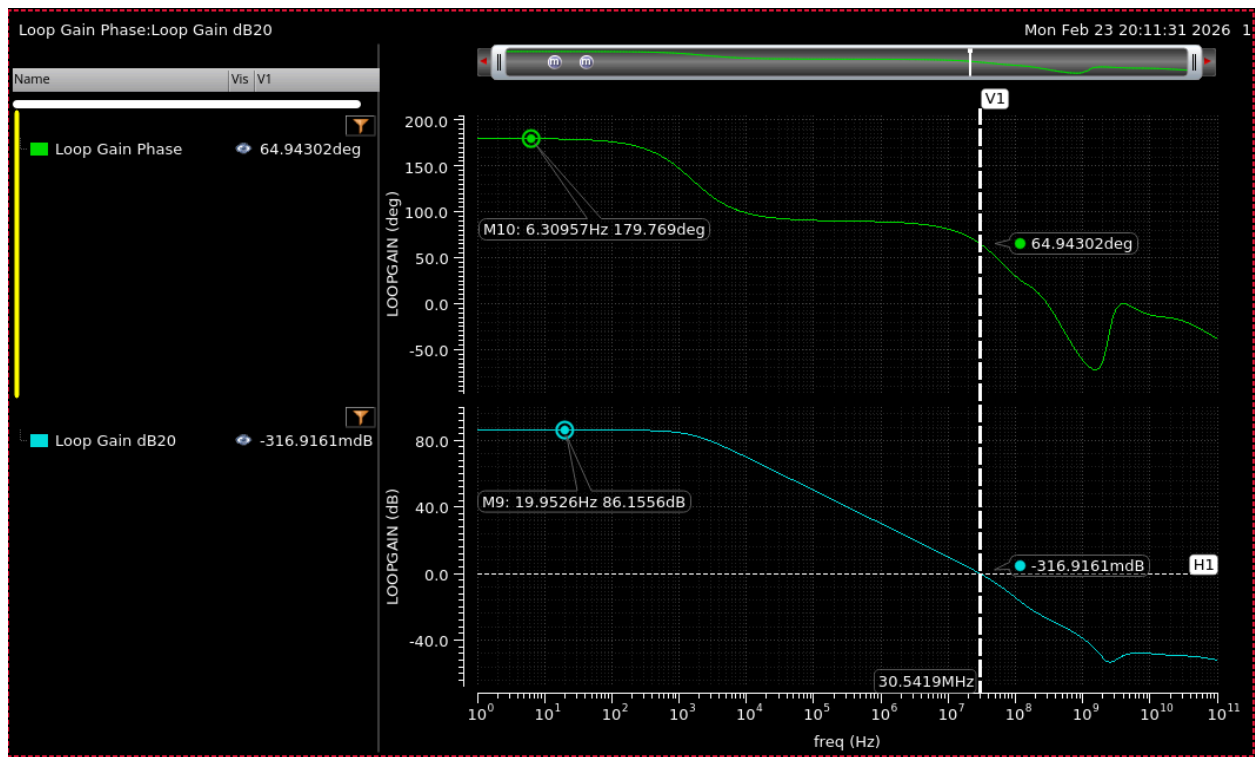
Values and Specs Being met:



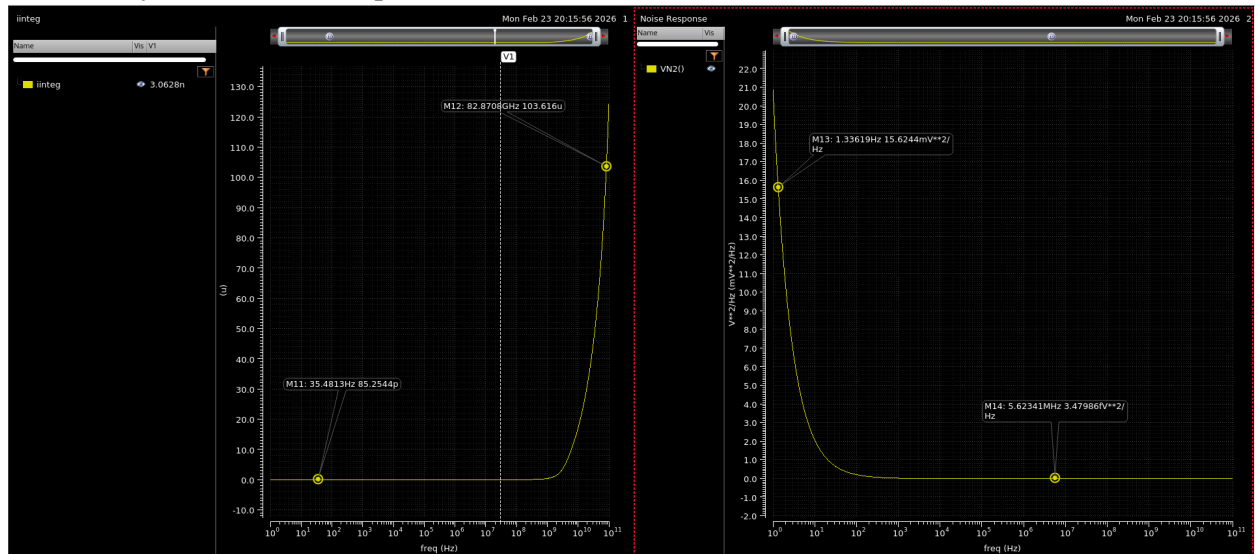
AC Simulation and Gain plots: Open loop Gain Plot:



Closed Loop Gain Plot:



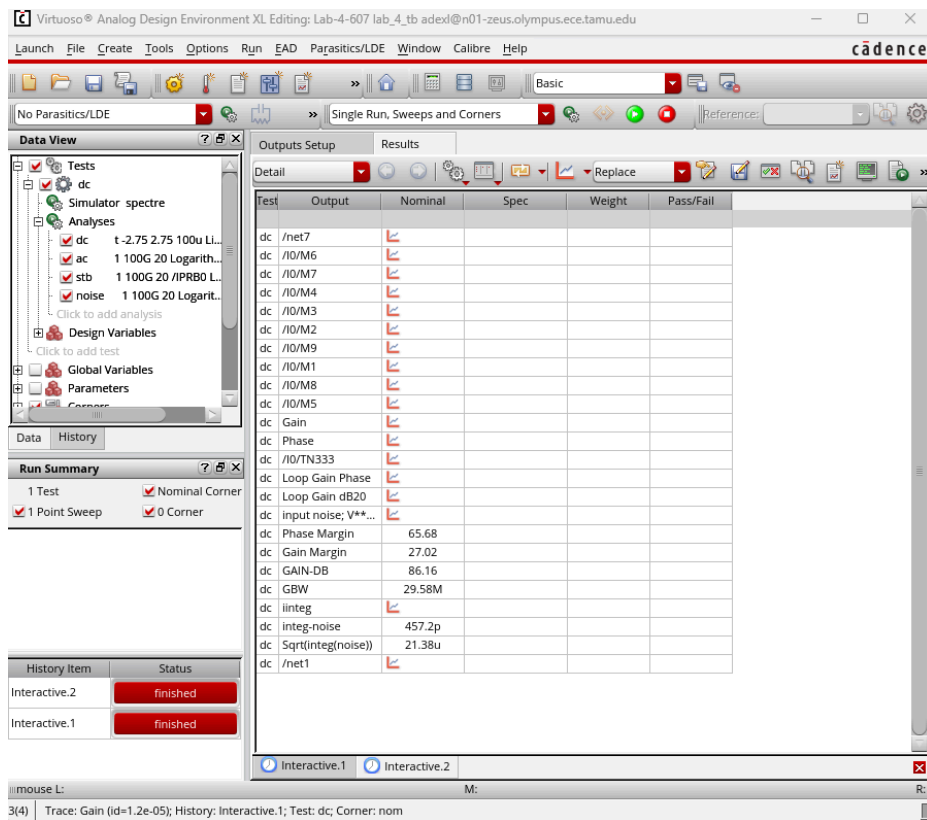
Noise Analysis and Noise Graphs:



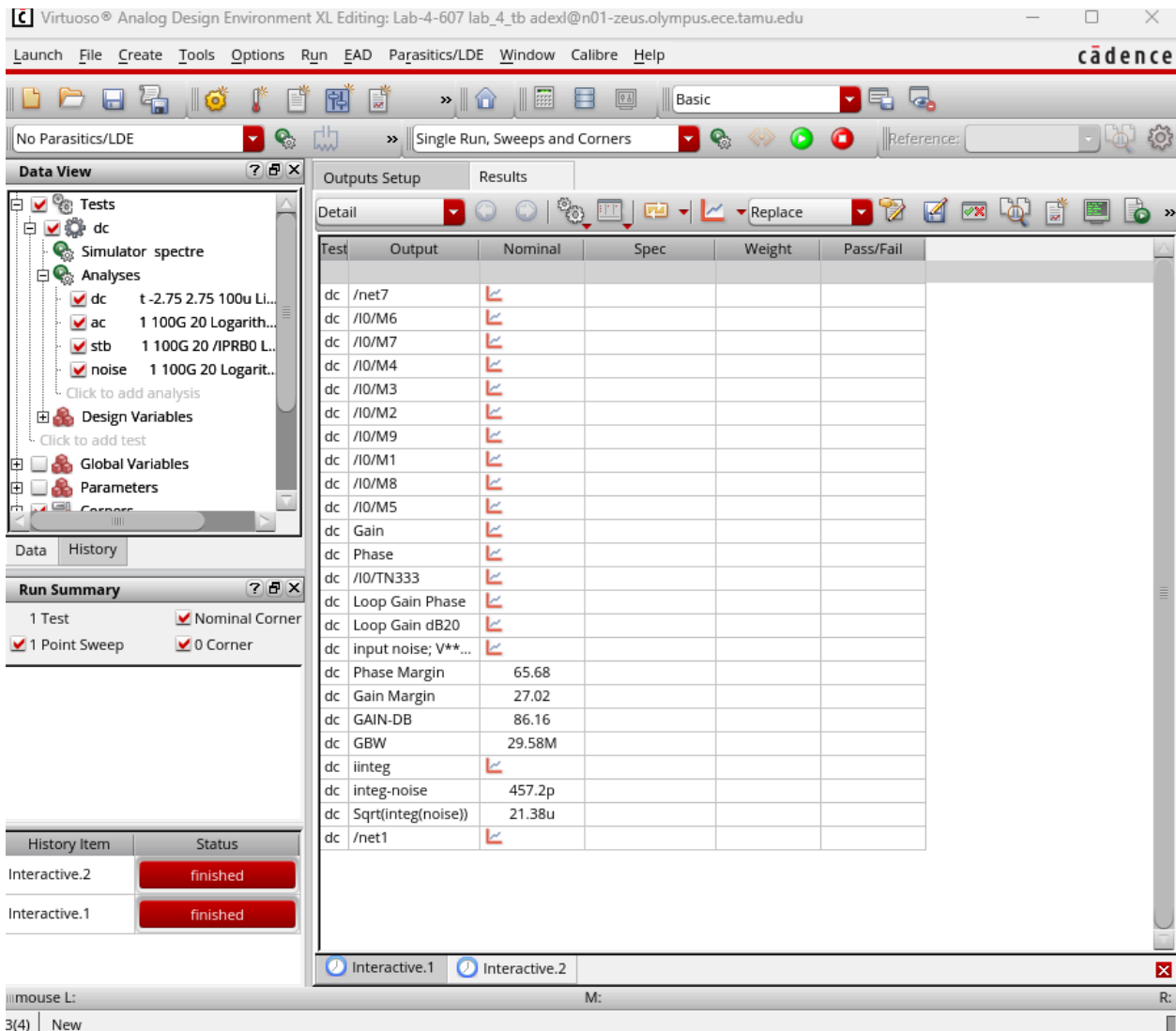
Values obtained by taking square root of the integrated value of noise graph:

12	integ-noise	457.2p
13	Sqrt(integ(noise))	21.38u

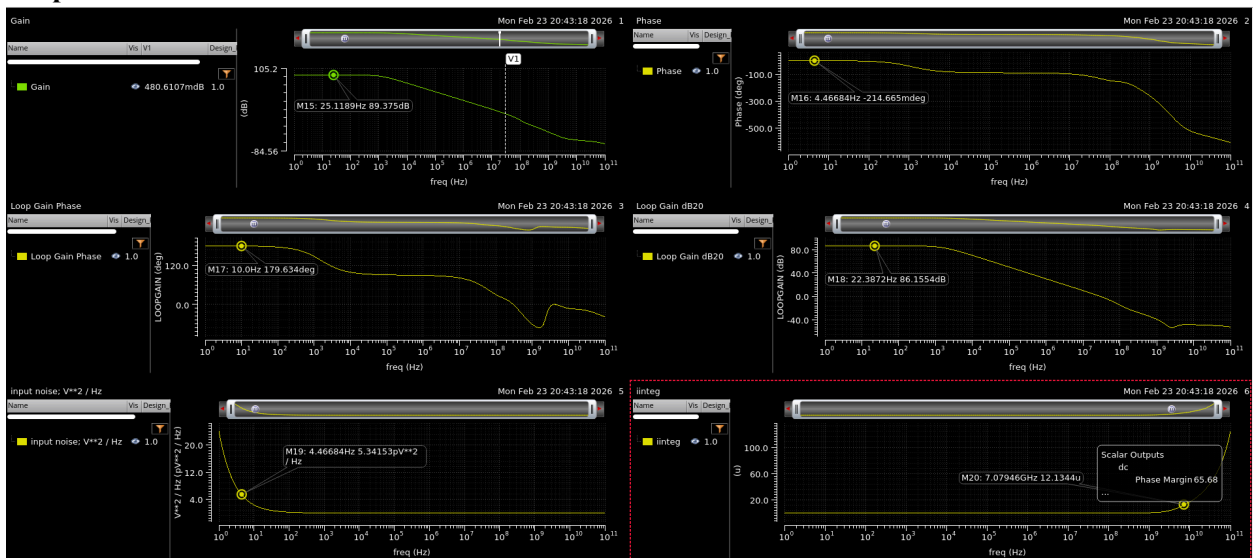
Value from ADEXL Window:



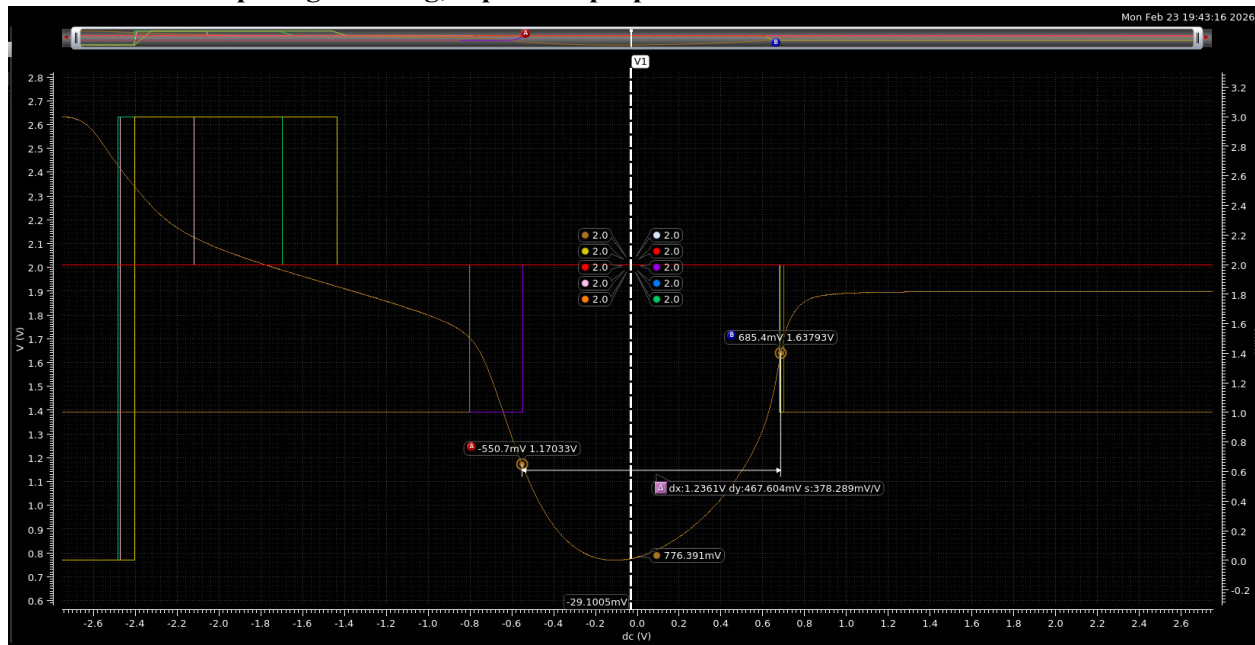
Graphs obtained from ADEXL:



Graphs From ADEXL Window:



2. Measure the output signal swing; report the proper screen shots.



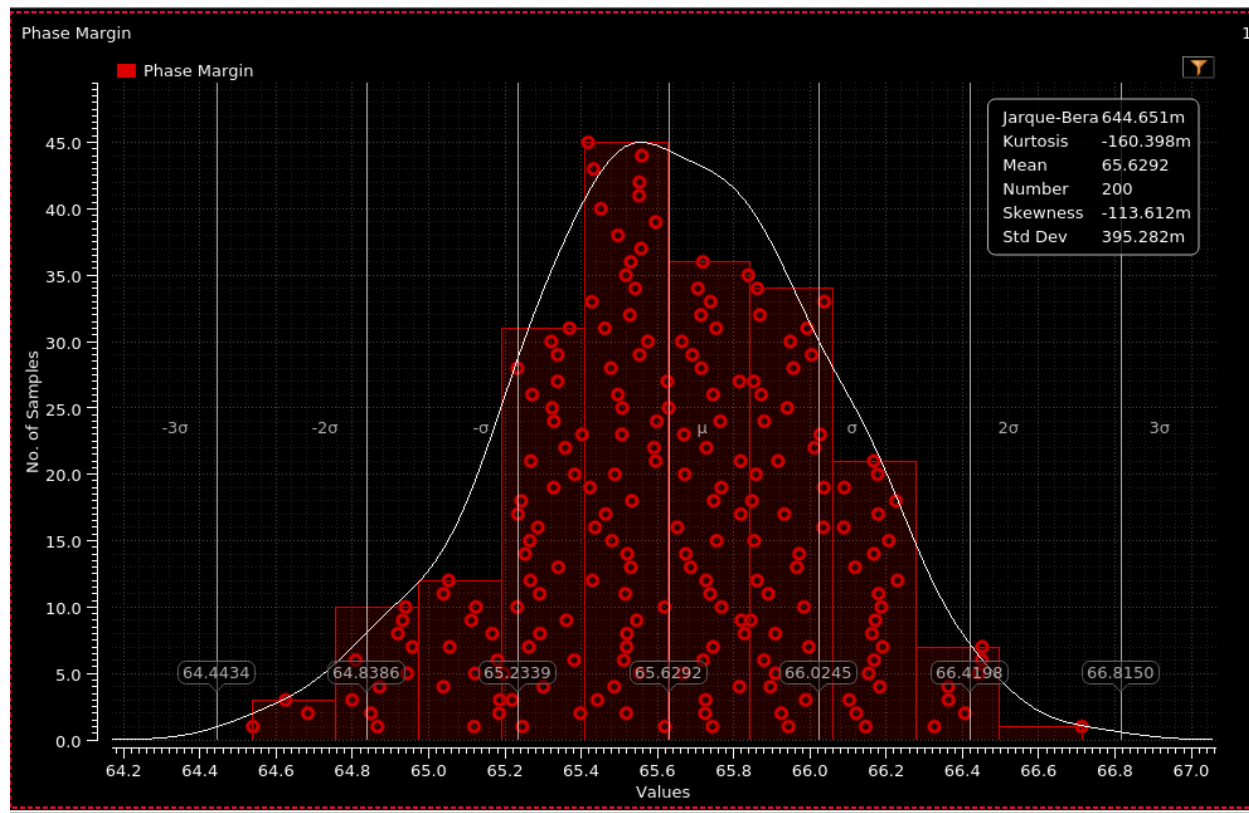
3. Do a cross simulation (supply variation and temperature variation) and report the worst case DC Gain, GBW, Phase Margin, Gain Margin and input referred noise. Assume there is +/- 5% variation on the supply, -40 <temperature <125 C. You do not have to adjust the design if the specs are not met. Running monte-carlo before running cross simulation:

Yield Estimate: 100 %(200 passed/200 pts) Confidence Level: <not set> Filter: <not set>

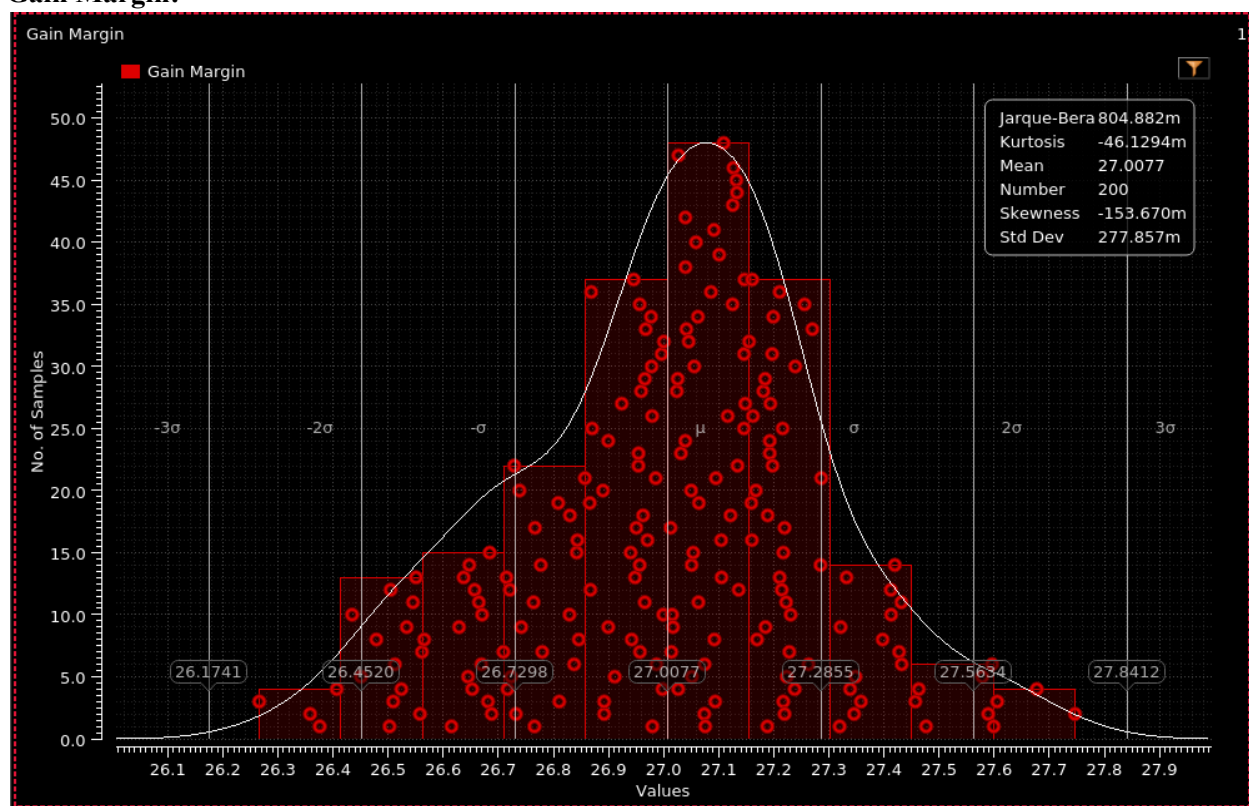
 dc

Phase Margin	100% (200/200)	64.54	info	66.71	65.63	395.3m	0
Gain Margin	100% (200/200)	26.27	info	27.75	27.01	277.9m	0
GAIN-DB	100% (200/200)	83.57	info	87.44	86.03	651.4m	0
GBW	100% (200/200)	28.18M	info	30.82M	29.6M	510.4K	0
integ-noise	100% (200/200)	215p	info	1.729n	505.7p	215.1p	0
Sqrt(integ(noise))	100% (200/200)	14.66u	info	41.58u	22.07u	4.308u	0

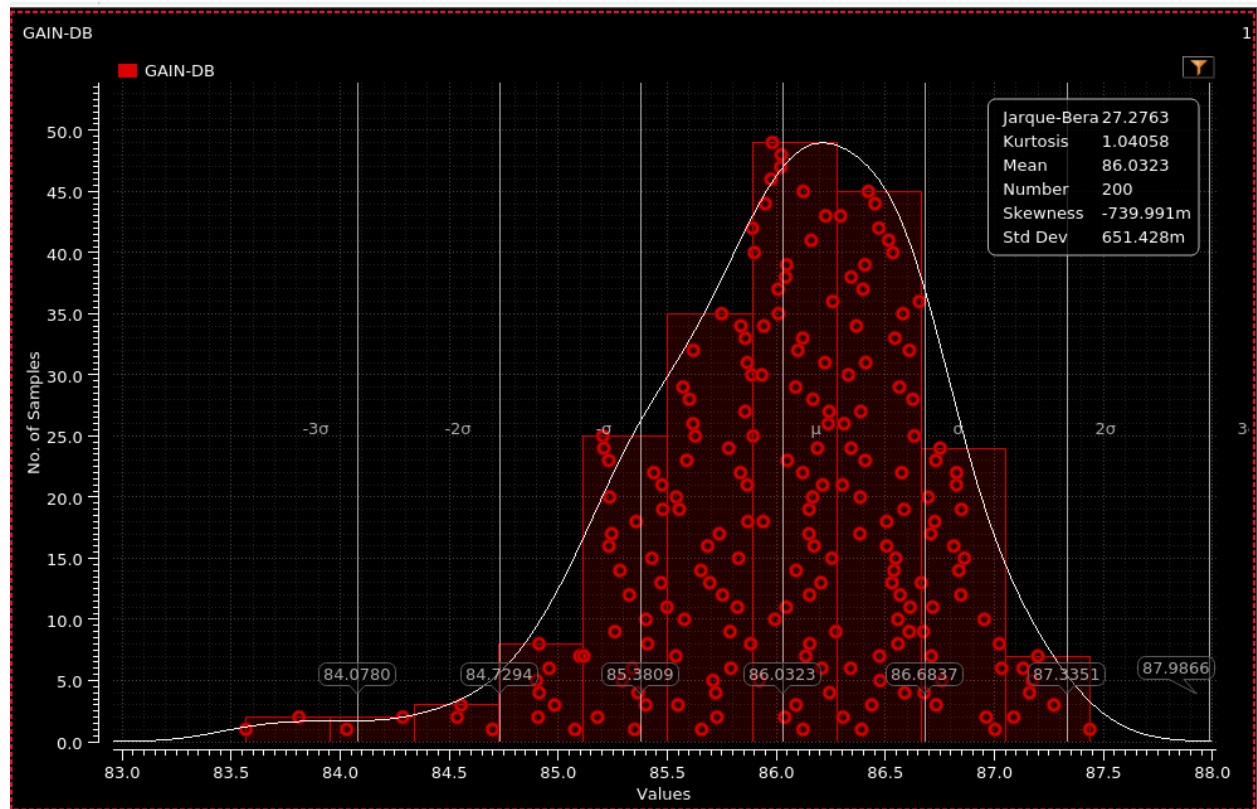
Phase Margin:



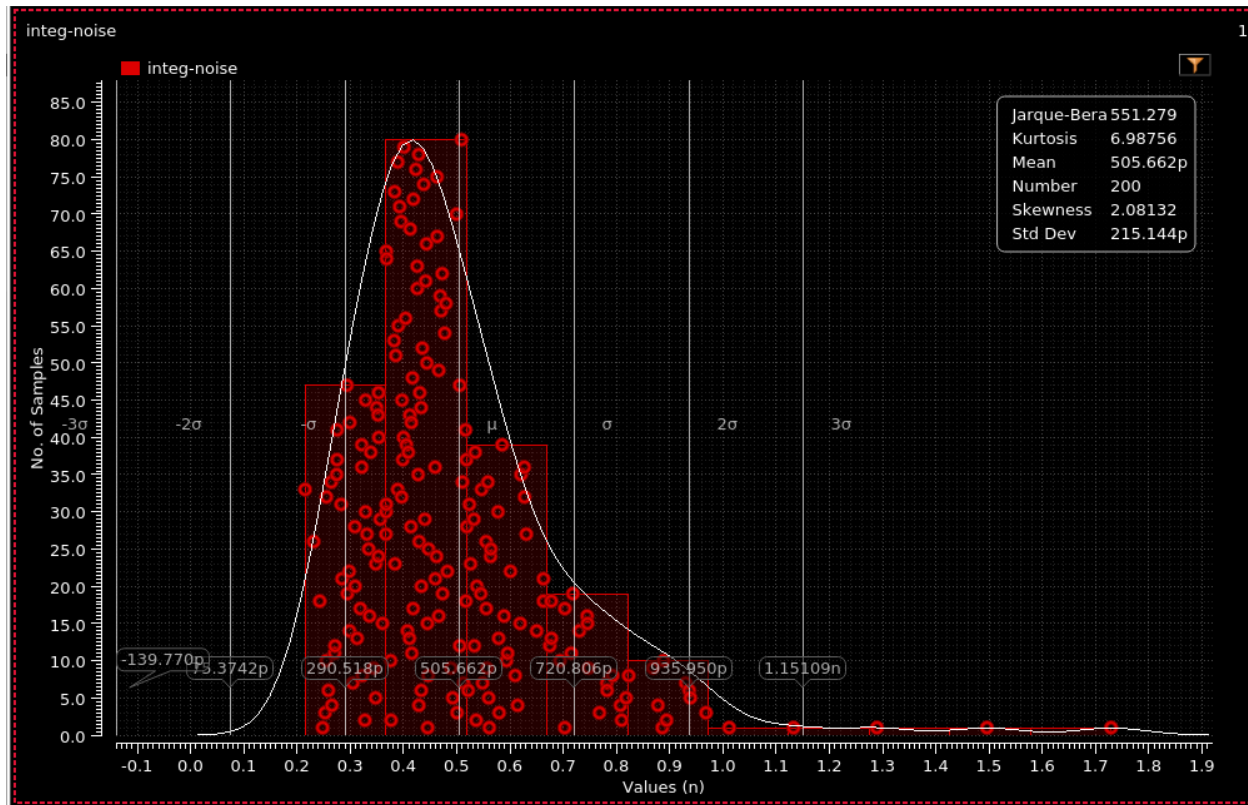
Gain Margin:



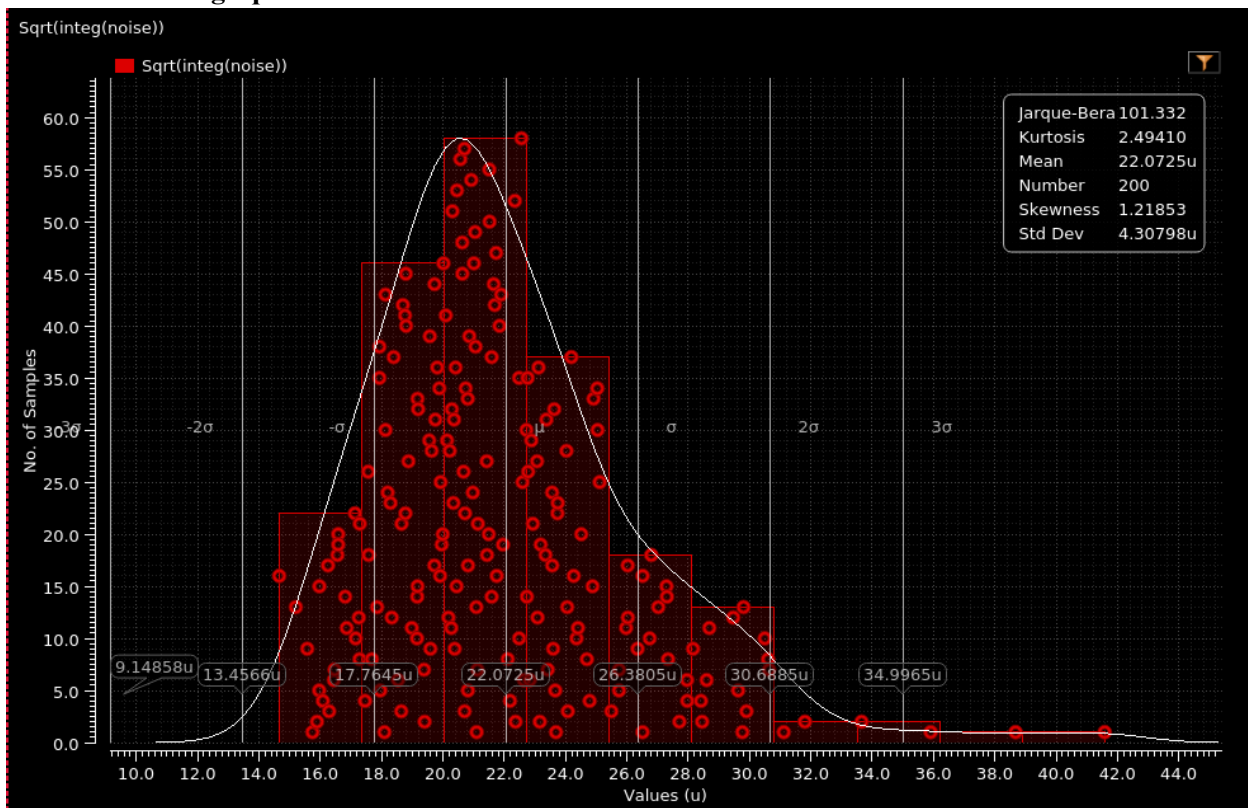
Gain-db:



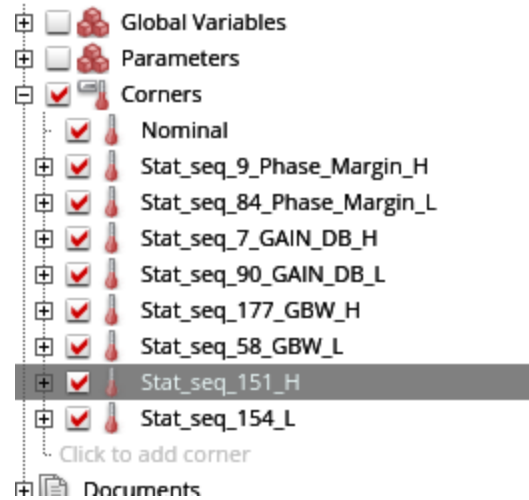
GBW:



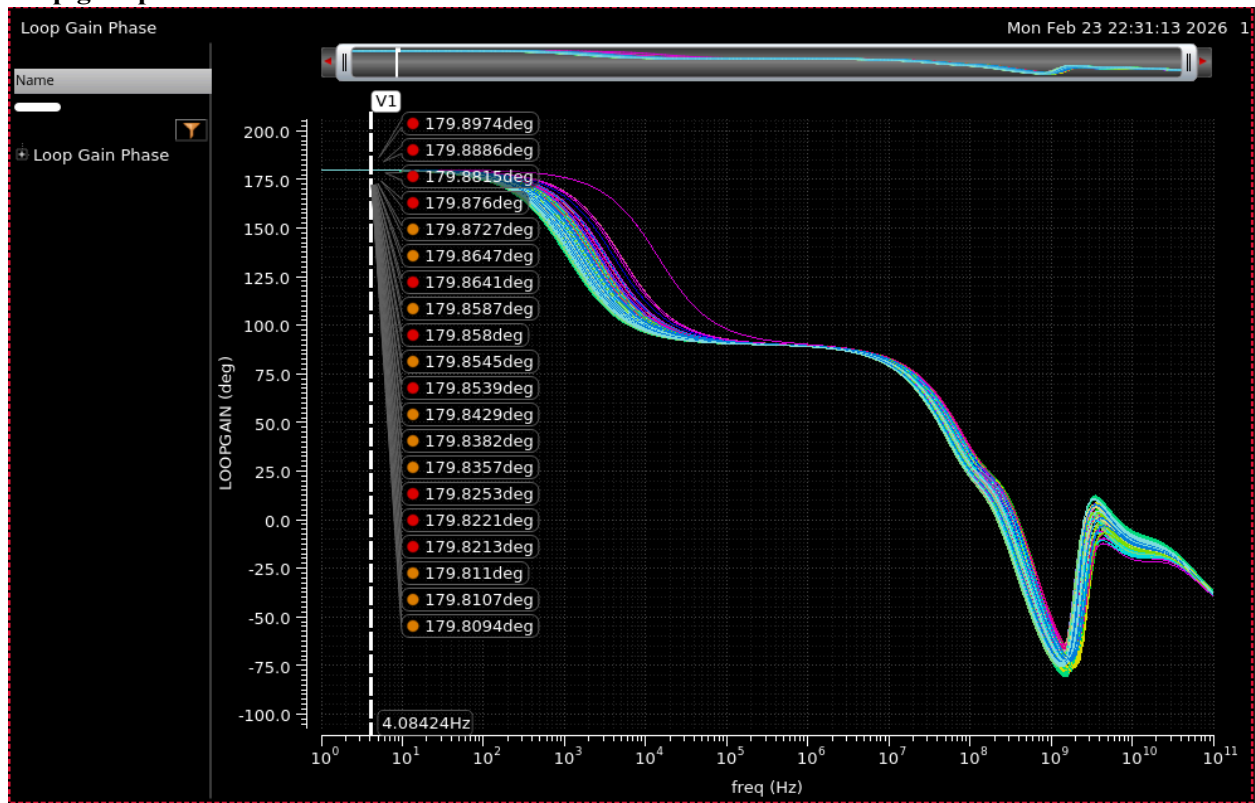
Noise after taking square root:



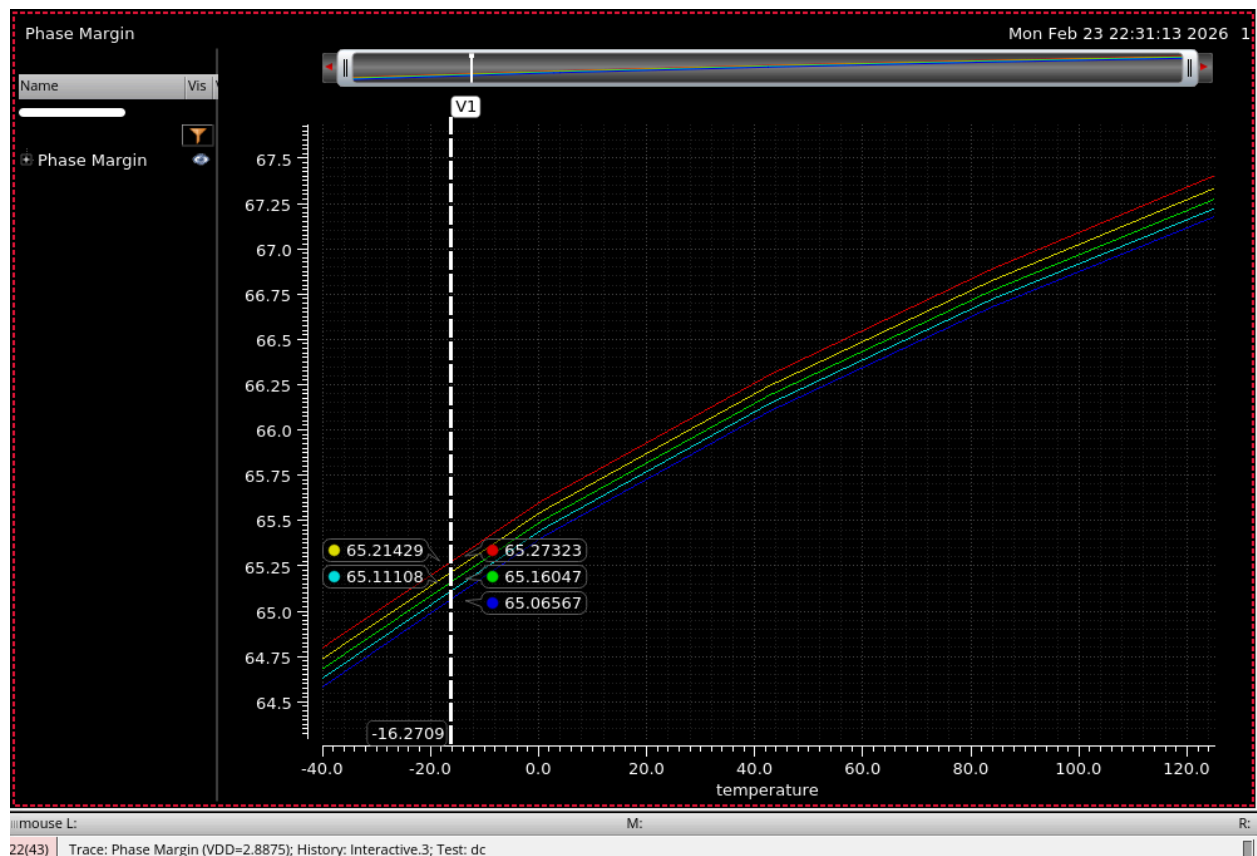
Creating Statistical Corners:



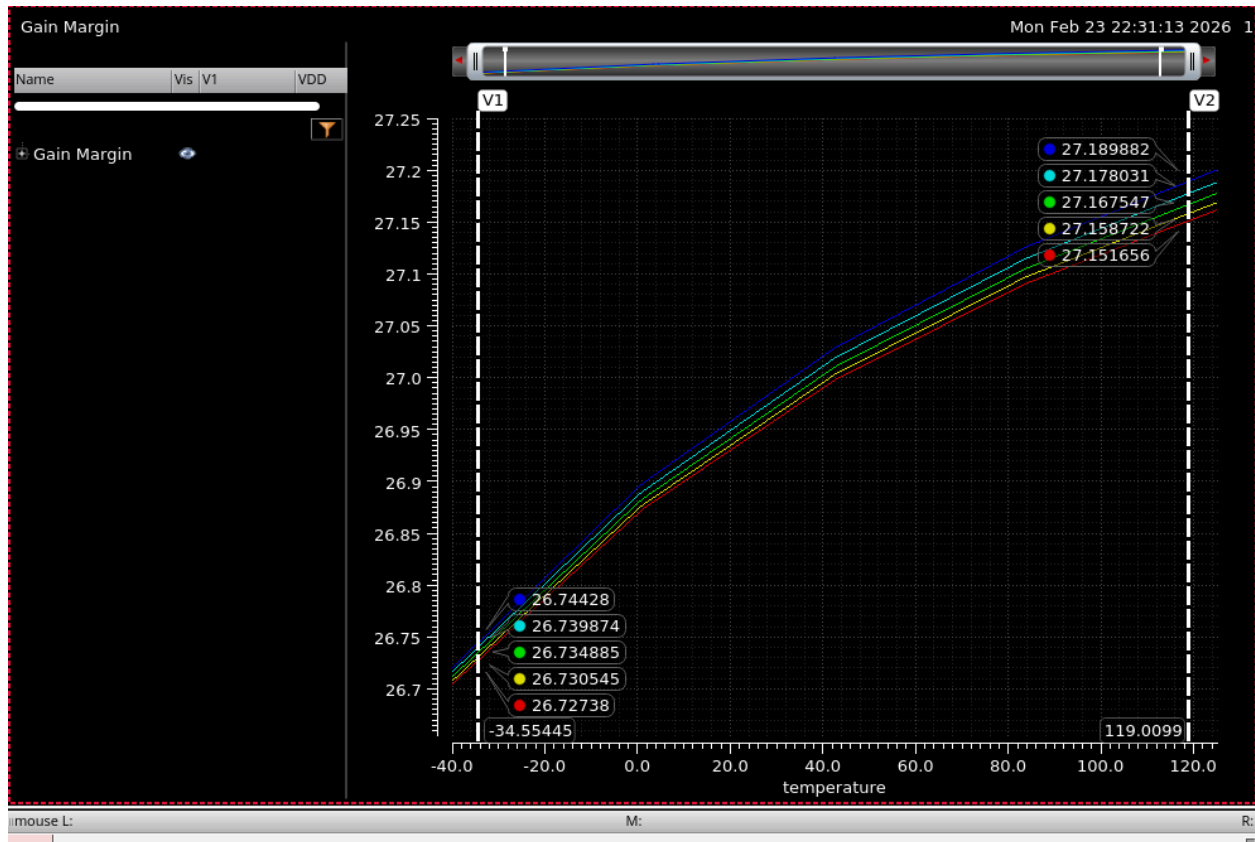
Loop gain phase:



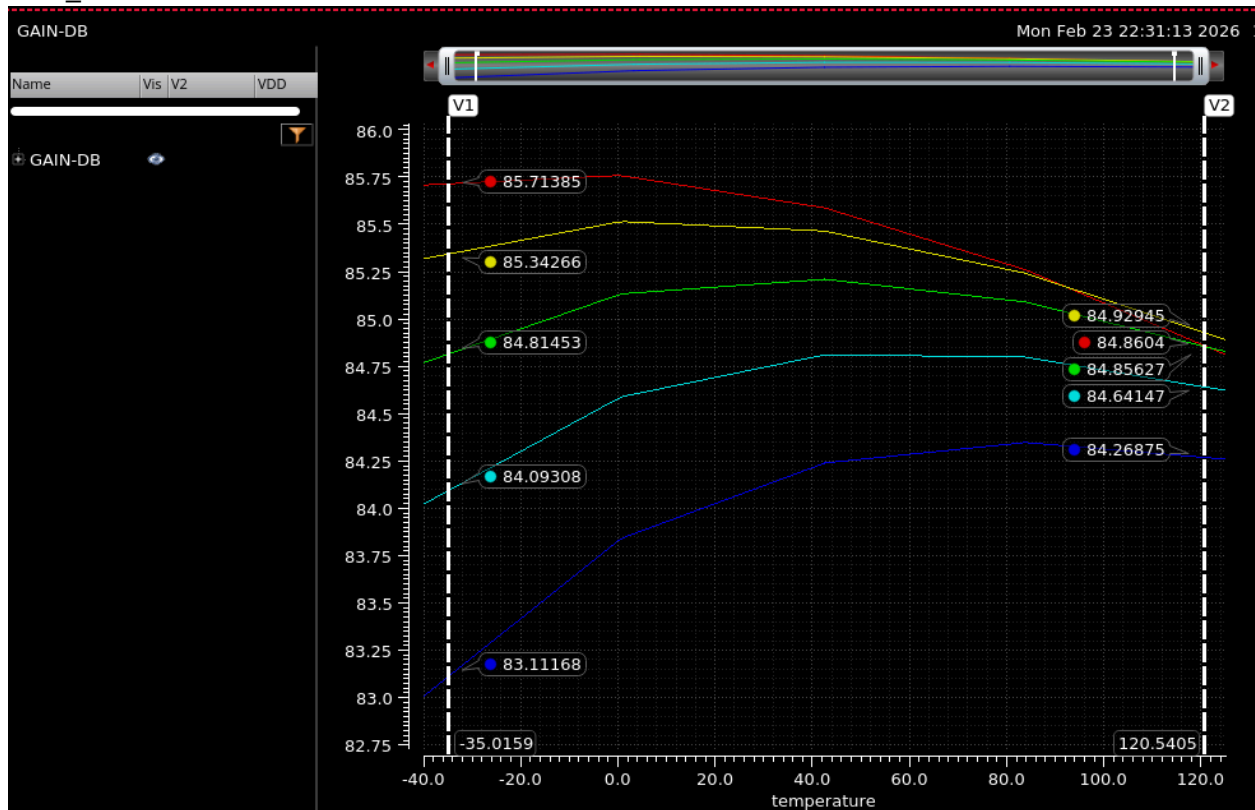
Gain db for all the corners:



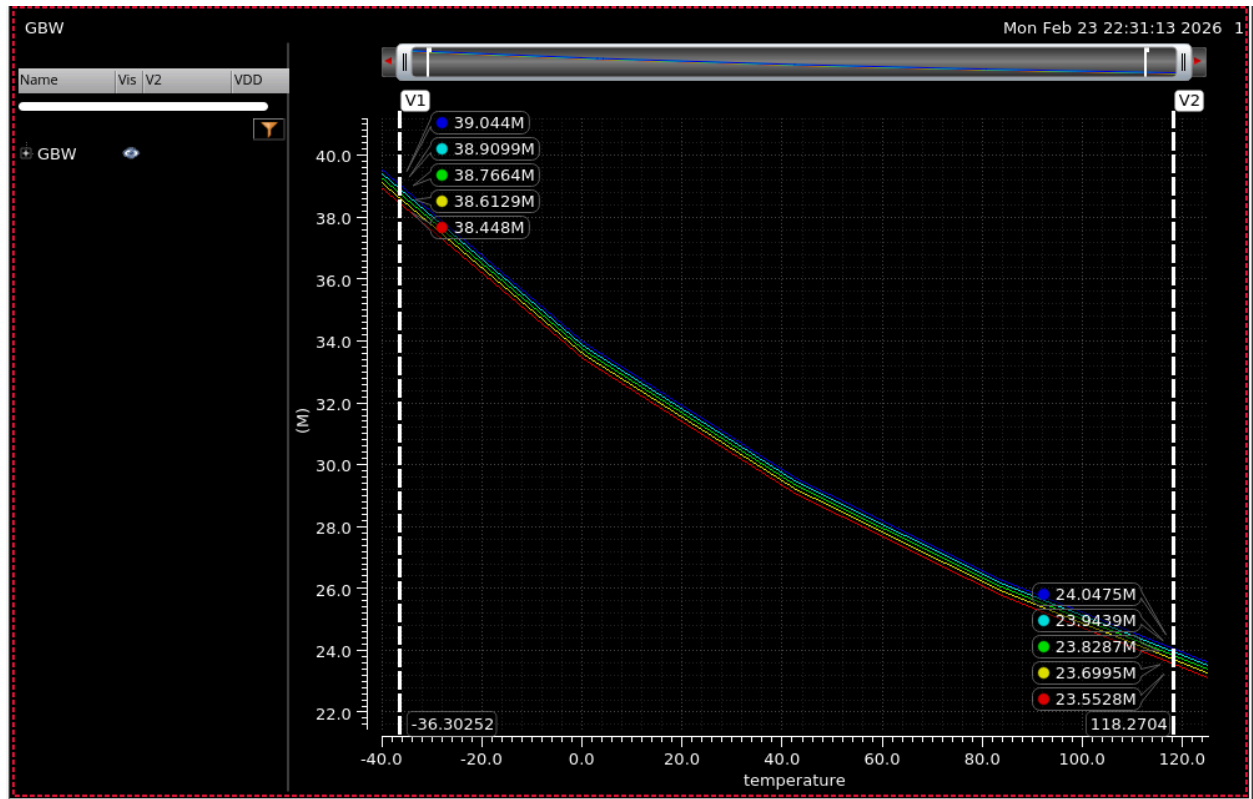
Gain_Margin:



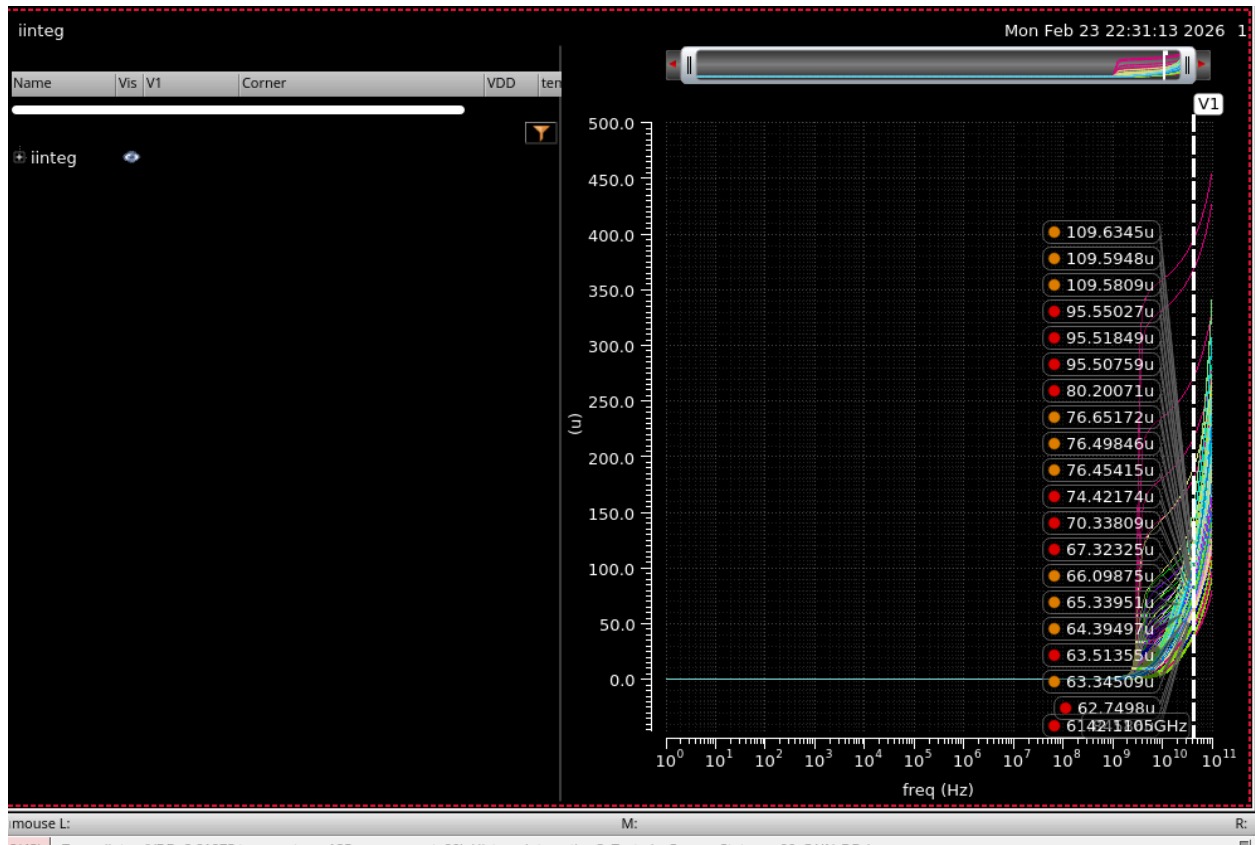
Gain_db:



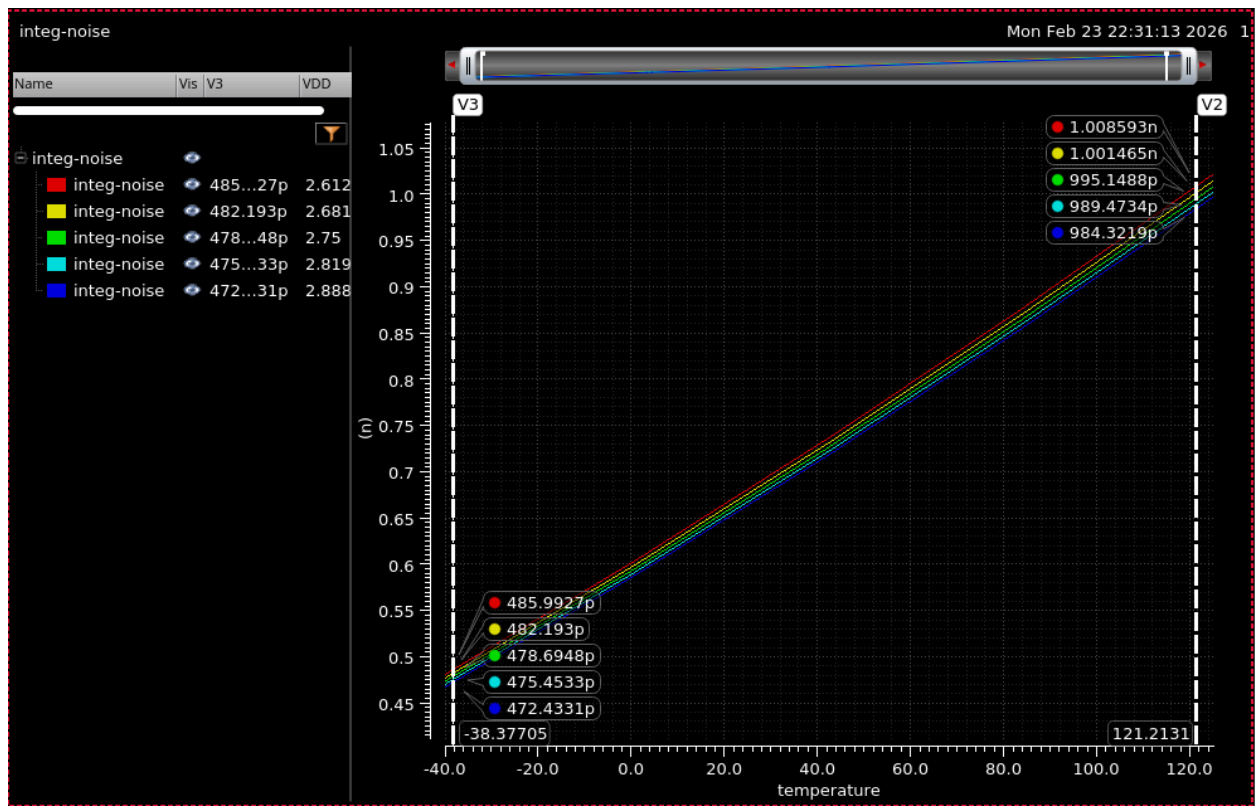
Gbw:



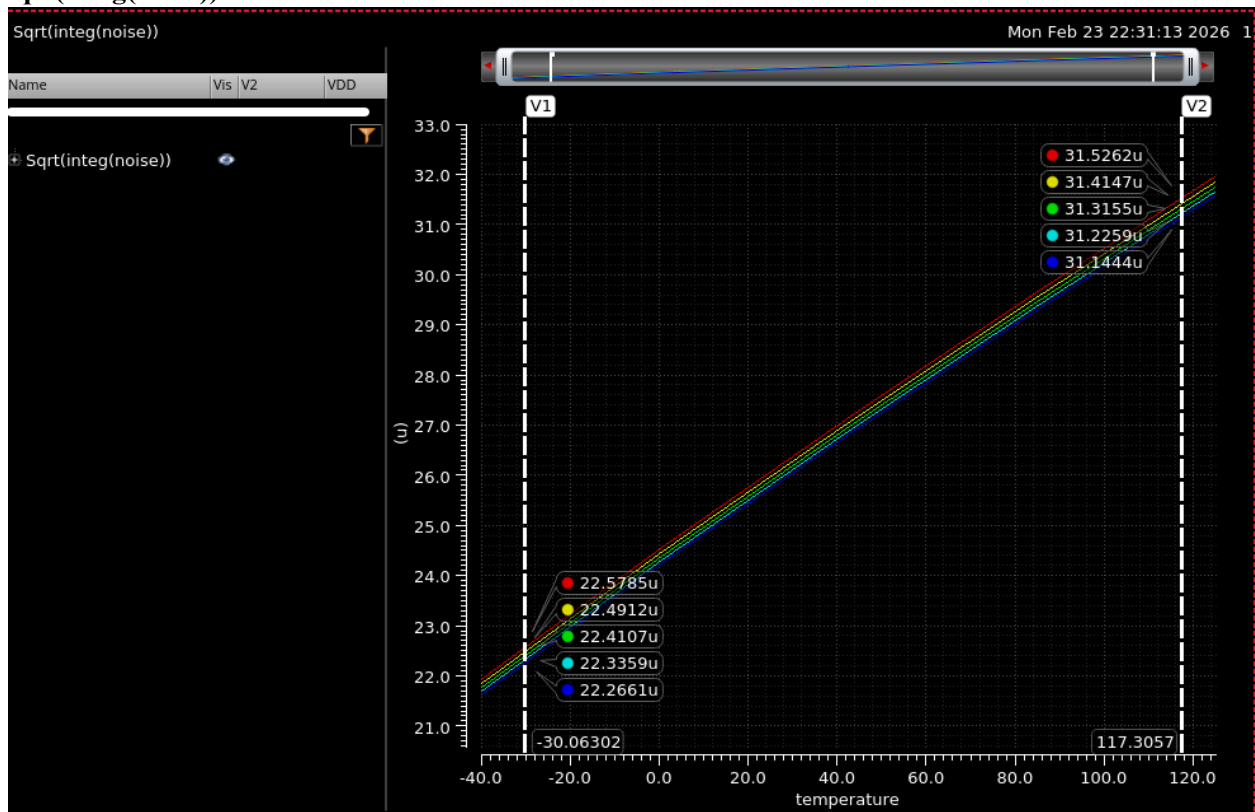
iinteg(Noise):



Integ_noise:



$\text{sqrt}(\text{integ}(\text{noise}))$:



Running Monte carlo simulation for

4. Provide comparison tables of hand-calculated vs. final transistor sizes, and required specs vs. simulated Specs.

Comparison table and Transistor Specs:

Transistor Number	Width	Handcalculated Width	Length	Hand Calculated Length
M1	100u	0.5u	1.4u	0.18u
M2	100u	0.5u	1.4u	0.18u
M3	100u	0.5u	1.4u	0.18u
M4	100u	0.5u	1.4u	0.18u
M5	30u	1u	1.4u	0.18u
M6	60u	0.75u	1.4u	0.18u
M7	60u	0.75u	1.4u	0.18u
M8	500n	0.65u	1.4u	0.18u
M9	500n	0.65u	1.4u	0.18u
BIAS NMOS	500n	- -	1.4u	- -
BIAS PMOS	3.6u	- -	1.4u	- -

Specs	Required Specs	Simulated Specs
Gain	>70 db	86.155 db
Phase Margin	>45 degree	65.68 degree
Output Swing	0.8V	1.4699 V
VDD-VSS	5.5V	5.5V
CL	2pF	2pF
Input Referred Noise	<30 u	21.38u
id	12u(calculated)	50u

5. Comment on your learning from this lab.

In this lab various simulations has been performed including dc,ac,stab and noise analysis and transient analysis:

This lab provided a comprehensive understanding of practical op-amp design beyond hand calculations, emphasizing iterative simulation-driven optimization.

One of the key learnings was how theoretical sizing from hand calculations significantly differs from final transistor dimensions required to meet all specifications simultaneously. Although initial widths and lengths were estimated analytically to meet gain and current targets, iterative refinement in Cadence was necessary to satisfy saturation conditions, gain, phase margin, noise, and output swing simultaneously.

A major insight from the design process was the importance of ensuring all transistors remain in saturation under DC operating conditions. During early simulations, some devices were entering the triode region. To resolve this, I systematically:

Increased channel length (L) to improve output resistance (r_o), thereby increasing gain and improving saturation margin.

Reduced bias current (I_B) strategically to reduce overdrive voltage (V_{ov}), which helped maintain proper headroom and ensured devices stayed in saturation.

Tuned W/L ratios iteratively to balance transconductance (g_m), gain, and noise performance.

This iterative adjustment of length, width, and bias current demonstrated the trade-off between gain, bandwidth, noise, and power consumption. Increasing length improved gain but reduced bandwidth; reducing current helped saturation but impacted GBW. The final design was achieved by carefully navigating these trade-offs.

The lab also reinforced the importance of different types of simulations:

1. DC Operating Point Analysis

Used to verify:

All transistors in saturation

Proper biasing (V_{B1} , V_{B2})

Correct current distribution

2. AC & STB Analysis

Verified open-loop gain (>70 dB achieved ~ 86 dB)

Confirmed phase margin ($\sim 65^\circ$)

Ensured stability using loop gain analysis

3. Output Swing Verification

Output swing was validated using transient analysis, where a large-signal input was applied and the maximum undistorted output range was measured. This demonstrated real large-signal behavior beyond small-signal AC assumptions.

4. Noise Analysis

Input-referred noise was obtained by integrating the noise spectral density and taking the square root of the integrated value. This highlighted the relationship between g_m and thermal noise ($4kT\gamma/g_m$), reinforcing how device sizing and bias current affect noise performance.

5. Monte Carlo and Corner Simulations

A critical learning component was performing:

Monte Carlo simulations to account for mismatch and process variations.

Supply variation ($\pm 5\%$ VDD) analysis.

Temperature sweep from -40°C to 125°C .

This helped evaluate worst-case:

DC gain

GBW

Phase margin

Gain margin

Input-referred noise

Running statistical corners demonstrated how performance shifts under real-world conditions and emphasized robust design rather than nominal-case optimization.

Key Takeaways

Op-amp design is an iterative optimization process, not a one-shot calculation.

Ensuring saturation across PVT variations is critical.

Increasing length improves gain (via r_o), while reducing bias current improves headroom but affects bandwidth.

Monte Carlo and cross-simulations are essential for verifying robustness.

Large-signal transient behavior (output swing) can differ significantly from small-signal AC results.

Trade-offs between gain, noise, GBW, and power are fundamental in analog design.

Overall, this lab strengthened my understanding of practical analog IC design methodology, simulation-driven refinement, and robust performance verification under process, voltage, and temperature variations.

